

Shazeb Ata

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Education

Arizona State University

Master of Science, Computer Science

August 2023 – May 2025

GPA: 4.0/4.0

- **Coursework:** Distributed Software Development, Distributed Database Systems, Cloud Computing

Indian Institute of Technology Roorkee

Bachelor of Technology, Engineering Physics

August 2016 – June 2020

GPA: 7.73/10.0

Experience

Software Developer – Data Science Team

Arizona State University

Tempe, Arizona

July 2025 – Present

- Built an interactive single-page React application utilizing functional components, useState for local state management, and useEffect for lifecycle handling and side effects
- Designed and implemented a RESTful API using FastAPI to process scheduling logic and return optimized task timelines based on inventory constraints
- Employed Docker for containerization, creating a multi-stage production-ready image combining the React frontend, MongoDB database and FastAPI backend into a single deployable unit

Senior Data Scientist (Backend Engineering)

HiLabs Inc

Pune, Maharashtra

October 2020 – May 2023

- Led a team in constructing a REST-API from scratch, leveraging Java Spring-Boot and deployed it on AWS
- Integrated API into Anthem's claims adjudication system, helping prevent \$2M+ in annual inpatient overpayments by streamlining inpatient data validation
- Authored Hive queries to ingest and explore claims data on Hadoop system, executed Spark Scala jobs via HiLabs MGest platform, and subsequently indexed data on Solr, greatly enhancing search capabilities
- Implemented DevOps pipelines utilizing services such as EC2, EMR, RDS, Load Balancer, and S3. Collaborated within the central log and disaster recovery team to ensure the resilience of the system and the integrity of the data

Projects

Aries Webpage

2019

- Developed RESTful routing via Express.js, where each route featured one or more handle functions and deployed the app on Heroku
- Established an inventory management system employing Node.js and MongoDB, with communication facilitated through Mongoose, and user authentication via Passport.js resulting in a 50% reduction in data entry errors

IoT Sensor Monitoring Platform

2025

- Built full-stack IoT monitoring system using React.js with WebSocket updates, Spring Boot REST API with Kafka consumer, Flask ML prediction service, PostgreSQL persistence, and event-driven microservices architecture achieving real-time data processing across 5+ devices
- Deployed cloud infrastructure using AWS CloudFormation to automate VPC, ECS Fargate containers, Application Load Balancer, RDS PostgreSQL database, and MSK Kafka messaging, with Docker Compose for containerization, Nginx for load balancing, and GitHub Actions CI/CD pipeline reducing deployment time by 70%
- Implemented real-time data pipeline using Apache Airflow for ETL automation, pandas for data manipulation, matplotlib for dashboard generation, and integrated ML prediction service with 95% accuracy, while creating responsive React dashboard with STOMP.js WebSocket protocol for sub-100ms UI updates

Scalable Multimedia Processing Pipeline On AWS

2024

- Orchestrated deployment of serverless components using AWS Lambda with Docker containers to process video frames via OpenCV and FFmpeg, then extended the system to EC2-based auto scaling using Amazon SQS to dynamically adjust worker nodes, reducing processing latency by 25% and cutting infrastructure costs

Skills

Languages: C++, Python, Java, Bash, Matlab, Scala, SQL, SAS, C#

Backend: Spring Boot, Next.js, Django, FastAPI, .NET, Node.js

Frontend: React.js, Tailwind.css, JavaScript, HTML/CSS

DevOps: Git, Microsoft Fabric, Linux, AWS, Docker

Big Data and Databases: MySQL, PostgreSQL, MongoDB, Snowflake, Solr, Hive, Hadoop, Apache Spark, Tableau